

# Calculus

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## HW2

65070021

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นายภิธานธิธาน กัณเฑาะฐมสิน

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Sec 1

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# Homework

65070021

นายภิตติคุณ ทัศนพรหมสิน

Find the derivative of the function below.

$$g(x) = (x^3 + 7x)(x + 3)$$

$$g'(x) = (x^3 + 7x)'(x + 3)' + (x + 3)(x^3 + 7x)'$$

$$= (x^3 + 7x) + (x + 3)(3x^2 + 7)$$

$$= x^3 + 7x + 3x^3 + 7x + 9x^2 + 21$$

$$= 4x^3 + 9x^2 + 14x + 21$$



$$f(x) = \frac{6x - 5}{x^2 + 1}$$

$$f'(x) = \frac{(x^2 + 1)(6x - 5)' - (6x - 5)(x^2 + 1)'}{(x^2 + 1)^2}$$

$$= \frac{(x^2 + 1)(6) - (6x - 5)(2x)}{(x^2 + 1)(x^2 + 1)}$$

$$= \frac{6x^2 + 6 - 12x^2 + 10x}{(x^2 + 1)^2}$$

$$= \frac{-6x^2 + 10x + 6}{(x^2 + 1)^2}$$



$$y = \frac{\sin x}{x^4}$$

$$y' = \frac{(x^4)(\sin x)' - (\sin x)(x^4)'}{(x^4)^2}$$

$$= \frac{(x^4)(\cos x) - (\sin x)(4x^3)}{(x^4)^2}$$

$$= \frac{x^{\cancel{3}}((\cos x)(x) - 4 \sin x)}{x^{\cancel{3}5}}$$

$$= \frac{x \cos(x) - 4 \sin(x)}{x^5}$$

